

CSA15 Cloud Computing and Big Data Analytics

Local IaaS Lab: Type-2 Hypervisor Installation and First Linux Virtual Machine

Course Code	CSA15	Course Title	Cloud Computing and Big Data Analytics
Guided Project	GP-VM-01	Submission	Individual evidence through GitHub and Google Classroom
Faculty	Dr C. Rajesh Babu	Employee ID	SSETSCS415
Target Learner	Beginner CSE student	Estimated Duration	3 to 4 lab hours
Primary Tool	Oracle VirtualBox or VMware Workstation + selected Linux ISO	Cloud Link	Models the local foundation of IaaS before public-cloud VM provisioning
Guided Project Pattern Read the idea, perform the action, observe the result, record evidence, explain what happened, extend the setup to your capstone context, and submit the final proof through GitHub and Google Classroom.			

Learning Outcomes

- Install either Oracle VirtualBox or VMware Workstation as a type-2 hypervisor.
- Select a guest operating system according to download size, laptop capacity and lab time.
- Create and configure a VM with CPU, RAM, virtual disk, storage controller and NAT network.
- Boot or install the selected guest OS and verify CPU, memory, storage and network evidence.
- Submit an individual evidence portfolio through GitHub and Google Classroom.

Before You Start

- Windows/macOS/Linux laptop with virtualization enabled in BIOS/UEFI or system settings.
- Minimum 8 GB host RAM recommended; keep at least 25 GB free disk space when using Ubuntu and at least 8 GB when using Tiny Core/Alpine.
- Administrator permission to install the hypervisor and its network drivers.
- One selected hypervisor and one selected guest OS image from the tables below.
- GitHub account and Google Classroom access.

Quick Learning Notes

- A host OS is the operating system already running on the laptop. A guest OS is the operating system running inside the VM.
- A type-2 hypervisor runs above the host OS, so it is suitable for student laptops and lab demonstrations.
- An ISO file behaves like a bootable installation DVD attached to the VM.
- NAT networking is preferred for this beginner lab because the guest VM can access the internet through the host without being directly exposed to the local network.
- The value 1 CPU, 2 GB RAM and 15 GB disk is the standard lab configuration; smaller OS images may safely use lower RAM/disk values if the student records the reason.

Lab Exercise Coverage

Lab Exercise	How This Guided Project Covers It
Experiment 7	Install a type-2 hypervisor, create and configure a VM image with a selected guest operating system.
Experiment 8	Create a VM with 1 CPU, 2 GB RAM and 15 GB storage disk, or record a justified lightweight-OS alternative.

Hypervisor Choice

Tool	When to Choose It	Student Evidence
Oracle VirtualBox	Choose when the laptop supports VirtualBox well, the student wants a free cross-platform tool, or VMware is not available.	VirtualBox version, New VM window, Settings screenshots and running guest OS proof.
VMware Workstation	Choose when VMware installer is available and the student wants to practice the lab-manual VMware workflow.	Installer file/version, VMware home screen, VM settings screenshots and running guest OS proof.

Guest OS Selection Guide

Choose one guest operating system based on laptop capacity, internet bandwidth and lab time. Record the selected image name, approximate size and reason for selection before creating the VM.

Option	Approx. Size	Best Use	Official Download	Suggested VM Setting
Tiny Core Linux Core	About 17 MB	Fastest boot and very low-bandwidth VM proof.	https://tinycorelinux.net/downloads.html	1 CPU, 256 to 512 MB RAM, 2 to 4 GB disk
Tiny Core Linux CorePlus	About 250 MB	Graphical desktop practice with small download size.	https://tinycorelinux.net/downloads.html	1 CPU, 512 MB RAM, 4 GB disk
Alpine Linux Standard	About 200 to 300 MB	Li3htwei3ht server-style Linux and packa3e practice.	https://alpinelinux.or3/downloads/	1 CPU, 512 MB to 1 GB RAM, 4 to 8 GB disk
Debian netinst	About 700 to 800 MB	Minimal standard Linux installation with optional packa3es.	https://www.debian.or3/distrib/netinst	1 CPU, 1 to 2 GB RAM, 8 to 15 GB disk
Ubuntu Server LTS	About 3 to 4 GB	Full Linux server practice when bandwidth is available.	https://ubuntu.com/download/server	1 CPU, 2 GB RAM, 15 GB disk
Ubuntu Desktop LTS	About 6 GB	Full desktop Linux practice when bandwidth and stora3e are available.	https://ubuntu.com/download/desktop	1 CPU, 2 to 4 GB RAM, 20 GB disk

Safety Rule

Do not allocate all host resources to the VM. Use NAT networking unless instructed otherwise. Never select a physical host disk for guest installation; the installation must happen only inside the virtual disk.

Architecture View

Laptop Hardware	Host OS	Selected Type-2 Hypervisor	Virtual Hardware	Guest OS ISO	Evidence
Laptop (Intel i3, 8 GB RAM, SSD)	Windows 10 pro Home 64-bit	Oracle VirtualBox 7.0.14	1 vCPU, 2 GB vRAM, 15 GB vDisk (VDI), NAT vNIC	Tiny Core Linux Core	Records 1-8 screenshots

The laptop remains the physical resource provider. The selected hypervisor creates virtual CPU, RAM, disk and network devices. The selected Linux image becomes the guest operating system.

Guided Build

Stage 1 - Confirm Host Readiness

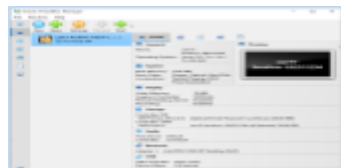
Learn: Virtualization support must be available before a VM can run reliably.

- Open Task Manager on Windows and check Performance -> CPU -> Virtualization. On macOS/Linux, use system information or terminal/system settings as available.
- Record host operating system, processor type, installed RAM and free disk space.
- Create a folder named CSA15_VM01_RegisterNumber on the host system.
- If virtualization is disabled, record the issue and enable it through BIOS/UEFI only with faculty or lab support.

Checkpoint

Host OS, RAM, free storage and virtualization status are recorded before downloading any installer.

Record 1 - Host readiness evidence



Stage 2 - Download the Selected Hypervisor

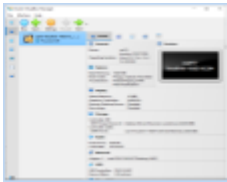
Learn: The hypervisor is the software layer that creates and manages the virtual computer.

- If choosing VirtualBox, open <https://www.virtualbox.org/wiki/Downloads> and select the platform package matching the host OS, such as Windows hosts or macOS hosts.
- If choosing VMware Workstation, use the faculty-shared installer or the official VMware/Broadcom Workstation download page and choose the installer matching the host OS.
- Save the installer inside the project folder or Downloads folder.
- Record installer filename, source link, download date and approximate file size.
- Do not download installers from unofficial software mirrors.

Checkpoint

The selected hypervisor installer file is available and its source is recorded.

Record 2 - Hypervisor download evidence



Stage 3 - Install the Hypervisor

Learn: During installation, the hypervisor adds drivers for virtual networking, USB integration and virtual hardware management.

- For VirtualBox, open the installer, proceed with default components, allow network interface installation when prompted, finish installation and open VirtualBox Manager.
- For VMware Workstation, open the installer, choose Typical setup, keep default installation path, allow network adapter installation, skip or disable optional customer-experience participation if shown, finish and reboot if requested.
- For VMware Workstation personal/free use prompts, select the free/non-commercial option when it is legally available for the installed edition.
- Open the installed application and capture the home screen or About screen.
- Record the exact installed version.

Checkpoint

The selected hypervisor opens successfully and the installed version is visible.

Record 3 - Hypervisor installation evidence

VirtualBox 7.0.14 installed with default components; network interface installation allowed. VirtualBox Manager opened; Help > About confirms version 7.0.14 r161095.

Stage 4 - Choose and Download the Guest OS

Learn: The guest OS image controls download time, VM size and installation complexity.

- Study the Guest OS Selection Guide table.
- Choose Tiny Core, Alpine, Debian netinst, Ubuntu Server or Ubuntu Desktop according to laptop capacity and internet speed.
- Download the ISO from the official link shown in the table.
- Record selected OS name, ISO filename, approximate download size and reason for choosing it.
- Keep the ISO file until the VM boots successfully and evidence is verified.

Checkpoint

The selected guest OS ISO is ready and the selection reason is written clearly.

Record 4 - Guest OS selection evidence

Guest OS: Ubuntu Server LTS 24.04. ISO: ubuntu-24.04-live-server-amd64.iso (~3.5 GB) from <https://ubuntu.com/download/server>. Reason: sufficient RAM/disk and stable broadband for realistic server/package practice.

Stage 5 - Create the VM Shell

Learn: A VM shell is the empty virtual computer before the guest OS starts.

- In VirtualBox, click New. In VMware, choose Create a New Virtual Machine.
- Name the VM CSA15-SelectedOS-RegisterNumber.
- Choose Linux as the OS family and the closest matching version. Use Other Linux when the exact lightweight OS is not listed.
- Store the VM inside a known folder, preferably CSA15_VM01_RegisterNumber.
- Attach the selected ISO during creation if the wizard allows it; otherwise attach it later through VM settings.

Checkpoint

The new VM appears in VirtualBox Manager or VMware Library with the correct name and OS family.

Record 5 - VM creation evidence



Stage 6 - Configure CPU, RAM, Disk and Network

Learn: CPU, RAM and disk values must be enough for the guest OS without damaging host performance.

- Set CPU to 1 processor/core for the lab.
- Set RAM from the OS table. For the standard lab configuration use 2048 MB.
- Create a dynamically allocated disk. Use 15 GB for the standard lab configuration or the OS-table value for lightweight OS choices.
- Use VDI for VirtualBox or VMDK for VMware unless the faculty specifies another disk format.
- Keep the network adapter as NAT.
- Open VM settings and capture CPU, RAM, disk and network screens before first boot.

Checkpoint

The VM settings show CPU, RAM, virtual disk size, disk format/controller and NAT network.

Record 6 - VM hardware configuration evidence

CPU: 1 processor. RAM: 2048 MB. Disk: 15 GB dynamically allocated VDI. Network: NAT. System/Storage/Network settings captured before first boot.

Stage 7 - Boot, Install or Run the Guest OS

Learn: The VM boots from the ISO first; after installation, it can boot from the virtual disk.

- Start or Power On the VM.
- If the OS is Tiny Core, reach the live desktop and record the desktop proof.
- If the OS is Alpine or Debian, reach the installer or shell/login prompt and follow the minimal installation path if instructed.
- If the OS is Ubuntu, choose Try/Install or Install Ubuntu according to lab time and faculty instruction.
- When an installer asks about disk usage, choose the VM virtual disk only. It is safe to erase the VM disk; it must not refer to the host physical disk.
- After boot or installation, capture the guest OS screen.

Checkpoint

The selected guest OS boots inside the VM and the evidence clearly shows it is running within the hypervisor window.

Record 7 - Guest OS boot or installation evidence

VM booted from ISO; selected Install Ubuntu Server; chose only the VM's 15 GB virtual disk (never host disk) when prompted. Installation completed to login prompt.

Stage 8 - Verify the VM from Inside the Guest OS

Learn: A configured VM is valid only when the guest OS can report its CPU, memory, storage and network state.

- Open terminal, shell or console in the guest OS.
- Run lscpu if available; otherwise record CPU details from the guest system or hypervisor settings.
- Run free -h or cat /proc/meminfo to verify memory.
- Run df -h to verify mounted storage.
- Run ip addr, ifconfig or ping 1.1.1.1 to verify networking.
- Capture all command outputs in a readable screenshot.

Checkpoint

CPU, RAM, disk and network evidence is captured from the guest OS or justified with available alternatives.

Record 8 - Guest verification evidence

lscpu: 1 CPU, x86_64. free -h: ~2.0 GiB RAM. df -h: 15 GB disk mounted at /. ip addr: NAT IP 10.0.2.15; ping 1.1.1.1 succeeded, confirming NAT internet access.

Final Evidence Checklist

Done	Checkpoint	Evidence to Capture
☐	Host readiness verified	Host OS, RAM, storage and virtualization status evidence.
☐	Hypervisor installed	VirtualBox or VMware installer/source/version and home screen evidence.
☐	OS choice justified	Selected OS, approximate ISO size, source link and reason for selection.
☐	VM configured	CPU, RAM, disk, disk format/controller and NAT network screenshots.
☐	Guest OS running	Boot/desktop/installer/login proof inside hypervisor window.
☐	Guest verification completed	CPU, RAM, disk and network command evidence.

Explain and Extend

Explain why VirtualBox or VMware is suitable as a type-2 hypervisor for this lab.

VirtualBox installs on top of the existing host OS, needs no dedicated hardware, is free for educational use, and its simple CPU/RAM/disk/NAT menus match this beginner lab exactly.

Explain why your selected OS is appropriate for your laptop and internet constraints.

Ubuntu Server LTS fits the 16 GB RAM / 120 GB disk laptop and available broadband for the ~3.5 GB ISO, and gives more realistic CLI/package/network practice than a minimal image.

Explain whether your VM configuration matches the standard 1 CPU, 2 GB RAM and 15 GB disk requirement or why you used a justified lightweight alternative.

The VM matches the standard specification exactly: 1 CPU, 2048 MB RAM, 15 GB VDI disk, since the laptop has ample spare capacity to support it without harming host performance.

Extend this setup to your capstone: what component could be tested in this VM?

A lightweight backend service or database (e.g. a Flask/Node API or PostgreSQL) can be tested here for the capstone, mirroring the move to a public-cloud IaaS VM later.

Submission Protocol

- Create a GitHub folder named CSA15-GP-VM01-Type2-Hypervisor-VM-Creation.
- Upload this completed document as DOCX/PDF along with screenshots and short README.md.
- README.md must include tool used, VM/cloud specifications, configuration decisions, screenshots index and cleanup status.
- Submit the GitHub repository link in Google Classroom. Do not upload passwords, private keys, tokens, public IP credentials or sensitive account screenshots.

Student Assurance

Declaration

I confirm that this guided-project evidence was completed by me, the screenshots are from my own lab environment, and no secret keys, passwords or sensitive account information are included.

Student Signature:
J Bharath

Register Number: 192511234

Date: